ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 4

Apply Naïve Bayes Classification Algorithms

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To implement Gaussian Naïve Bayes for binary classification of the Diabetes dataset and evaluate its performance using standard classification metrics.

2. Objectives

- To convert the continuous target into two classification classes.
- To understand the probabilistic basis of Naïve Bayes.
- To train a Gaussian Naïve Bayes classifier.
- To evaluate accuracy, precision, recall and F1-score.
- To analyze predictions using a confusion matrix.

3. Dataset Description

The Scikit-learn Diabetes dataset contains 442 observations and 10 independent features: age, sex, bmi, bp, s1, s2, s3, s4, s5 and s6. For regression experiments, the continuous target is used directly. For classification experiments, the continuous target was converted into two classes using the median value of 140.5: Lower Progression and Higher Progression.

Property	Value
Samples	442
Features	10
Classification classes	2
Median threshold	140.5
Train/Test split	80% / 20%

4. Theory

Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It estimates the probability that an observation belongs to a class given its observed features and selects the class with the highest posterior probability.

Bayes' theorem is expressed as $P(C|X) = P(X|C)P(C) / P(X)$, where C represents a class and X represents the observed feature vector. $P(C|X)$ is the posterior probability, $P(X|C)$ is the likelihood, $P(C)$ is the prior probability and $P(X)$ is the evidence.

The 'naïve' assumption is that features are conditionally independent given the class. This assumption is often not perfectly true in real datasets, but the algorithm can still perform effectively and is computationally efficient.

Gaussian Naïve Bayes is appropriate when continuous features are assumed to follow a Gaussian or normal distribution within each class. For every feature and class, the algorithm estimates a mean and variance and uses these values to calculate likelihoods.

The Gaussian probability density is proportional to $\exp(-(x-\mu)^2/(2\sigma^2))$, where μ is the class-specific mean and σ^2 is the class-specific variance. These probability estimates are combined with class priors to obtain the posterior classification probabilities.

In this experiment, the original target is continuous, so it is converted into a binary class using the median target value of 140.5. Values at or above the median are placed in Higher Progression and the remaining values are placed in Lower Progression.

Classification performance is evaluated using accuracy, precision, recall and F1-score. Accuracy measures the overall proportion of correct predictions. Precision measures the proportion of predicted positives that are actually positive, while recall measures the proportion of actual positives that are correctly identified.

The F1-score is the harmonic mean of precision and recall. It is useful when both types of error matter and provides a balanced summary of the two measures.

The confusion matrix gives a detailed view of correct and incorrect predictions. It contains true positives, true negatives, false positives and false negatives, allowing class-specific performance to be examined.

Naïve Bayes is known for being relatively fast because training mainly involves estimating probability distributions from the training data rather than solving a complex optimization problem. It can work well even with relatively small datasets.

One limitation is the conditional-independence assumption. When features are strongly related, the probability estimates may not perfectly represent the real joint distribution. Nevertheless, the classifier can remain useful because classification decisions depend on the relative posterior probabilities.

The train-test split is used to estimate how the trained model performs on unseen observations. In this experiment, 80% of the data is used for training and 20% for testing. The test set is not used to estimate the model parameters.

Because the classification threshold is derived from the dataset median, the two classes represent lower and higher progression relative to the sample distribution rather than a clinical diagnosis.

5. Methodology

- Load the Diabetes dataset.
- Use the median target value, 140.5, to create two classes.
- Split the data into training and testing sets using an 80:20 ratio.
- Train a Gaussian Naïve Bayes classifier using the ten independent features.
- Predict the classes of the test observations.
- Calculate accuracy, precision, recall and F1-score.
- Generate a confusion matrix and class-distribution visualization.

6. Outputs

6.1 Classification Report

Unnamed: 0	precision	recall	f1-score	support
Lower Progression	0.7619047619047619	0.7111111111111111	0.735632183908046	45.0
Higher Progression	0.723404255319149	0.7727272727272727	0.7472527472527473	44.0
accuracy	0.7415730337078652	0.7415730337078652	0.7415730337078652	0.7415730337078652
macro avg	0.7426545086119554	0.7419191919191919	0.7414424655803966	89.0
weighted avg	0.7428708035927734	0.7415730337078652	0.7413771815166622	89.0

6.2 Confusion Matrix Values

Unnamed: 0	Predicted Lower	Predicted Higher
Actual Lower	32	13
Actual Higher	10	34

6.3 Class Distribution

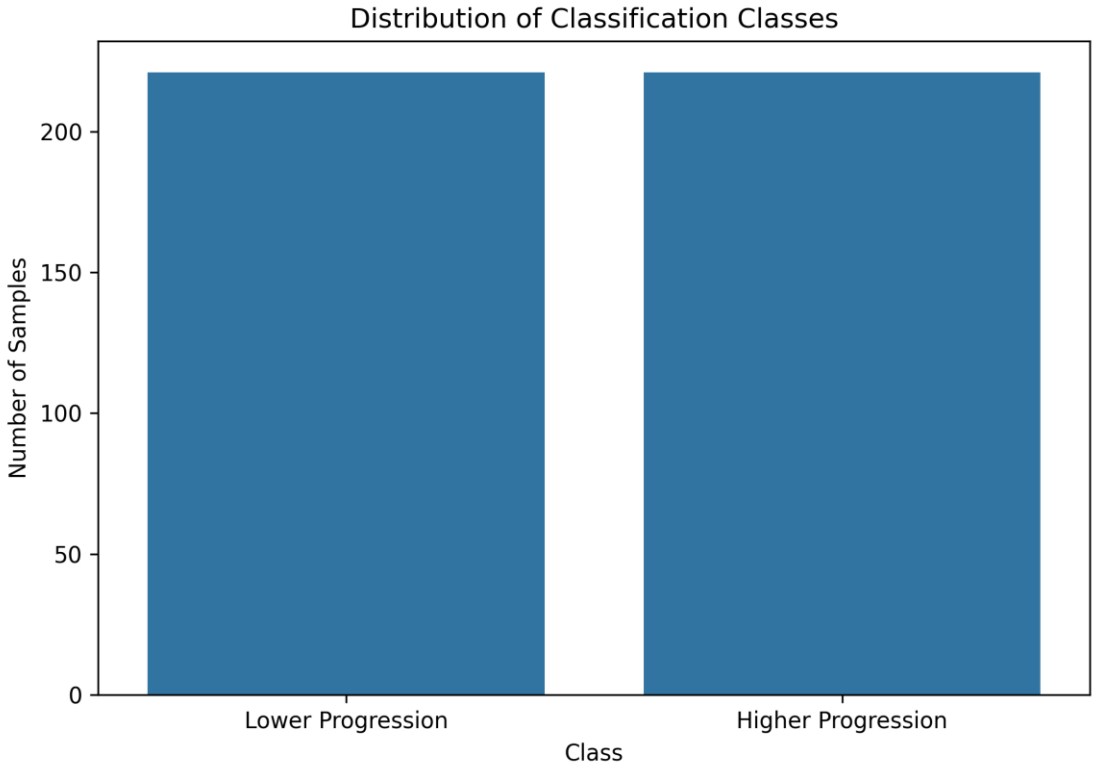


Figure 1: Distribution of lower and higher progression classes

6.4 Confusion Matrix

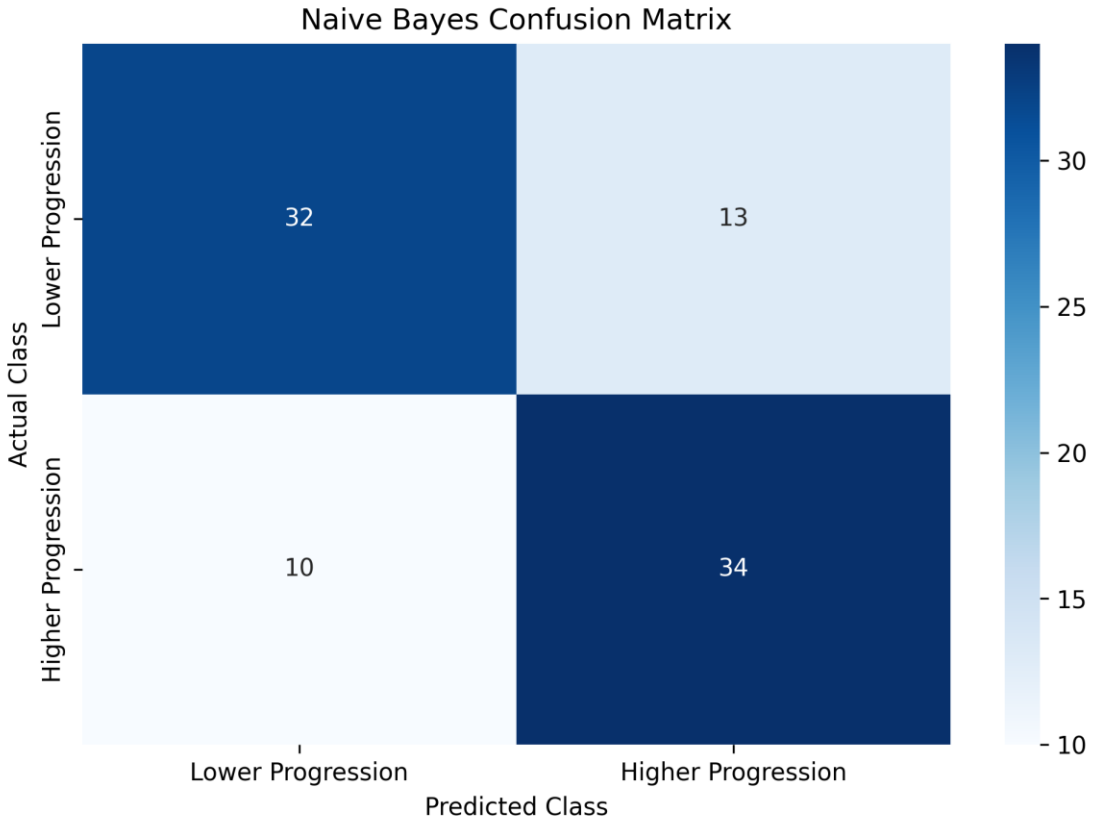


Figure 2: Gaussian Naïve Bayes confusion matrix

7. Result

Gaussian Naïve Bayes was successfully implemented. The model achieved an accuracy of 74.16% on the 89 test samples. For Lower Progression, precision was 0.7619 and recall was 0.7111. For Higher Progression, precision was 0.7234 and recall was 0.7727.

8. Observations

- The target was converted into two classes using the median value 140.5.
- The model achieved 74.16% accuracy.
- Higher Progression recall was 0.7727, meaning most higher-progression test cases were identified.
- The confusion matrix shows 32 correctly classified Lower Progression samples and 34 correctly classified Higher Progression samples.
- Precision and recall provide more detailed information than accuracy alone.

9. Conclusion

The experiment demonstrates the application of Gaussian Naïve Bayes to a binary classification problem. The probabilistic classifier successfully separated the two progression categories and was evaluated using multiple metrics and a confusion matrix.